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B.Tech. Degree IV Semester Examination April 2019

ME 406 MANUFACTURING PROCESS (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer ALL questions)

(8 × 5 = 40)

- I. (a) What are the applications of casting?
- (b) Write a note on slag trap system.
- (c) What are the features to aid handling in a casting operation?
- (d) Write a note on shell moulding.
- (e) Describe the nature of plastic deformation in metal forming process.
- (f) What are the different roll stand arrangements?
- (g) Compare the three flames in gas welding.
- (h) Write a note on explosion welding.



PART B

(4 × 15 = 60)

- II. (a) What are the different types of patterns? (7)
- (b) Describe mechanized moulding process. (8)
- OR**
- III. (a) What are the different types of gates in casting? (7)
- (b) What are the tests conducted to ensure the quality of sand mould? (8)
- OR**
- IV. (a) Explain casting metallurgy. (7)
- (b) Explain centrifugal casting with neat sketches. (8)
- OR**
- V. (a) Discuss the design considerations in casting process. (7)
- (b) Describe the equipment used in die casting. (8)
- OR**
- VI. (a) What are the different types of roll passes? (7)
- (b) Compare drop and press forging. (8)
- OR**
- VII. (a) Explain wire drawing. (7)
- (b) Describe the various sheet metal operations. (8)
- OR**
- VIII. (a) Explain the mechanism of metal transfer in arc welding. (7)
- (b) What are the different kinds of electrodes used in arc welding? (8)
- OR**
- IX. (a) Discuss the heat input at different zones of a weld during the welding process. (7)
- (b) Describe diffusion welding and electro slag welding. (8)